Introduction to Innovation & Design Thinking

This session aims to guide you through a structured process for ideating innovative solutions. Empowering the next generation of tech entrepreneurs to ideate, build and scale capstone projects into sustainable startups.

by Inbaraj Suppiah

The Trainer

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Tech entrepreneur, community builder and innovation strategist dedicated to shaping Malaysia’s digital and startup ecosystem. Through JomHack, Malaysia’s leading corporate innovation platform, he drives impactful programmes that foster collaboration, digital transformation and emerging tech adoption. In partnership with Gamuda Berhad, he founded the Gamuda AI Academy to upskill Malaysians in AI and data technologies. Passionate about empowering talent, Inbaraj continues to champion innovation that connects creativity, technology and human potential.

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What is Innovation?

Innovation is the process of introducing new ideas, methods or products to create value, solve problems or improve existing solutions.

Key Elements of Innovation:

Problem-Solving

Addressing unmet needs or inefficiencies that are serious pain points to a large market segment.

Creativity

Generating original and imaginative ideas to solve high-pain problems that you're passionate about.

Value Creation

Delivering measurable benefits that can improve processes, products, services or experiences.

Implementation

Turning great innovative ideas and plans into actionable, executable solutions on the ground.

Types of Innovation

Product Innovation

Developing new or improved products that can create a new way of life
(e.g. Smartphones, LLMs, AI tools)

Process Innovation

Enhancing methods for efficiency or increasing productivity
(e.g. factory automation, sales funnel)

Business Model

Introducing new ways of delivering value to users and generating revenue
(e.g. SaaS, API as Service)

Social Innovation

Solving societal challenges to improve way of life and the living environment
(e.g. Renewable Energy, EdTech)

Characteristics of Innovation

Disruptive Innovation

Breaks traditional norms and introduces groundbreaking changes to an industry or ecosystem.

Incremental Innovation

Focuses on gradual improvements to existing systems to avoid major disruptions to an industry.

Sustainable Model

Prioritises long-term benefits to company and users while addressing environmental or societal issues.

User-Centric Innovation

Always aims to meet the needs and preferences of users or consumers first before revenue & profits.

Why Innovation Matters?

Competitive Edge

Helps organisations stay ahead in rapidly changing markets.

Economic Growth

Drives industries and creates new opportunities.

Social Impact

Improves quality of life and addresses global challenges.

Innovation is not just about inventions; it’s about applying ideas effectively to make a meaningful impact.

What is Design Thinking?

WHY DOES IT MATTER?

“ Design thinking is a structured human-centered approach for creative problem-solving, involving five key stages: empathise, define, ideate, prototype and test. This approach prioritises user needs and rapid iteration to develop innovative solutions. “

Design Thinking Principles

1 Empathise

Understanding the user's needs, pain points, perspectives and emotions through research.

2 Define

Clearly articulating the pain points you want to address into a human-centered problem statement.

3 Ideate

Generating a wide range of potential solutions through brainstorming and creative techniques.

4 Prototype

Quickly building low-fidelity prototypes to visualise, test and refine ideas, digitally or in physical form.

5 Test It

Validate assumptions based on user feedback and iterating on solutions based on real-world insights.

Understand the Problem:

User Research & Insights

1 User Interviews

Conducting in-depth interviews with potential or intended users to gather their needs and challenges.

2 Surveys

Distributing surveys to collect quantitative data and identify user preferences and pain points.

3 Observation

Observing users in their natural environment to understand their patterns, behaviours and interactions to gauge their needs.

AI-Centric Empathy

In AI projects, create empathy through understanding the "Shadow Problem" that users can't articulate but data reveals.

- Data Empathy: Analysing existing logs for friction points.
- User Intent: What do they WANT vs. What do they ASK for?
- Trust Barriers: Identifying where users fear AI bias or complexity.

Rapid Ideation Techniques:

Brainstorming

Generating a large number of ideas without judgement, encouraging creativity and collaboration from members.

Crazy 8s

A rapid sketching technique where participants create multiple sketches in a short time - divergent thinking.

Brainwriting

A method that encourages equal participation from team members, reduces bias and diverse idea generation quickly.

Basic Brainstorming

THE OG METHOD

A **basic brainstorming method** is a straightforward approach to idea generation that focuses on gathering as many ideas as possible within a set timeframe. It's great for fostering creativity, encouraging participation and identifying potential solutions to a problem.

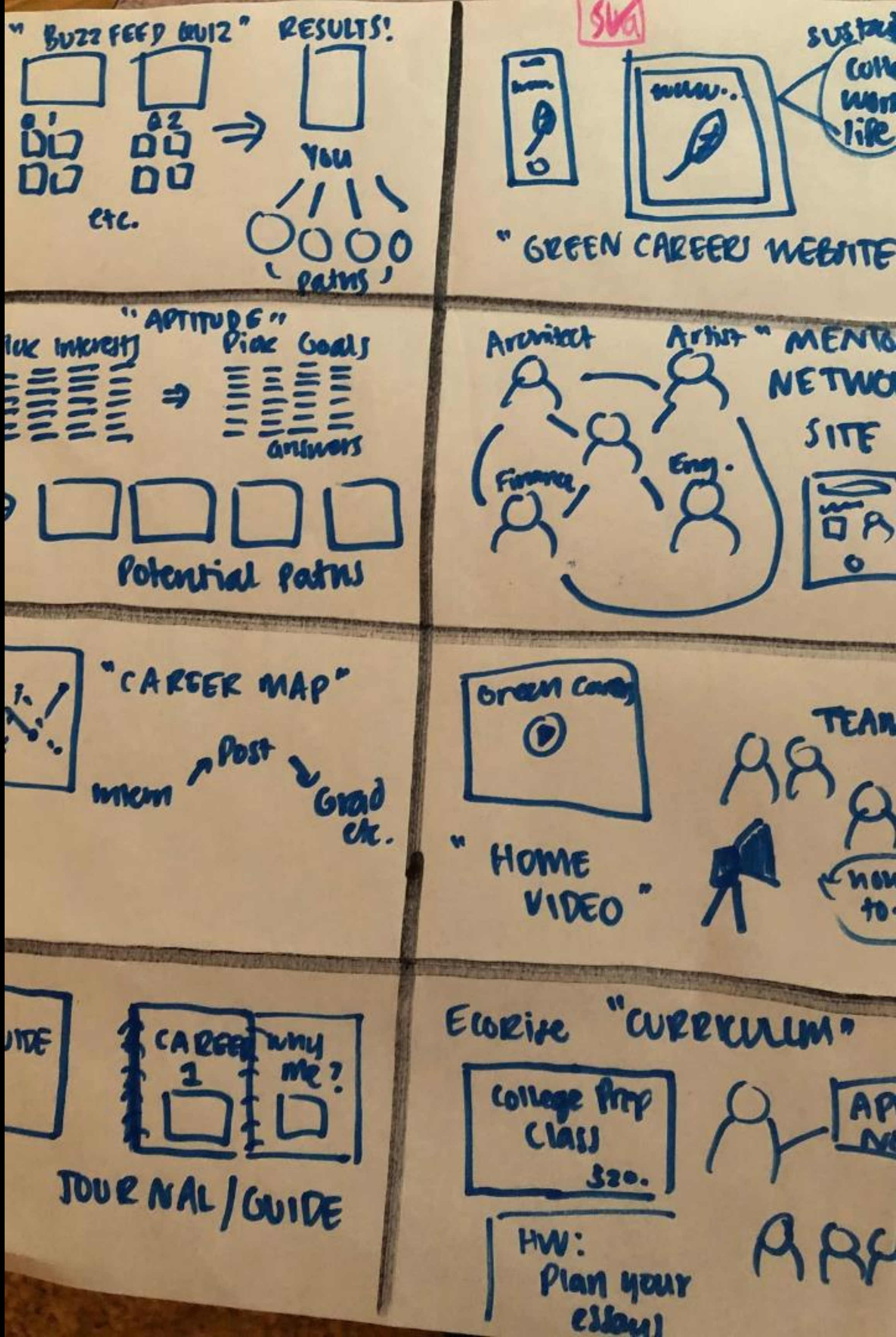
Brainstorming Method:
Step-by-Step

1. Define the Problem or Goal
2. Set the Rules
3. Gather the Team
4. Set Time Limit
5. Idea Generation
6. Encourage Participation
7. Evaluate & Prioritise
8. Action Plan

Crazy 8s

8 SKETCHES, 8 MINUTES.

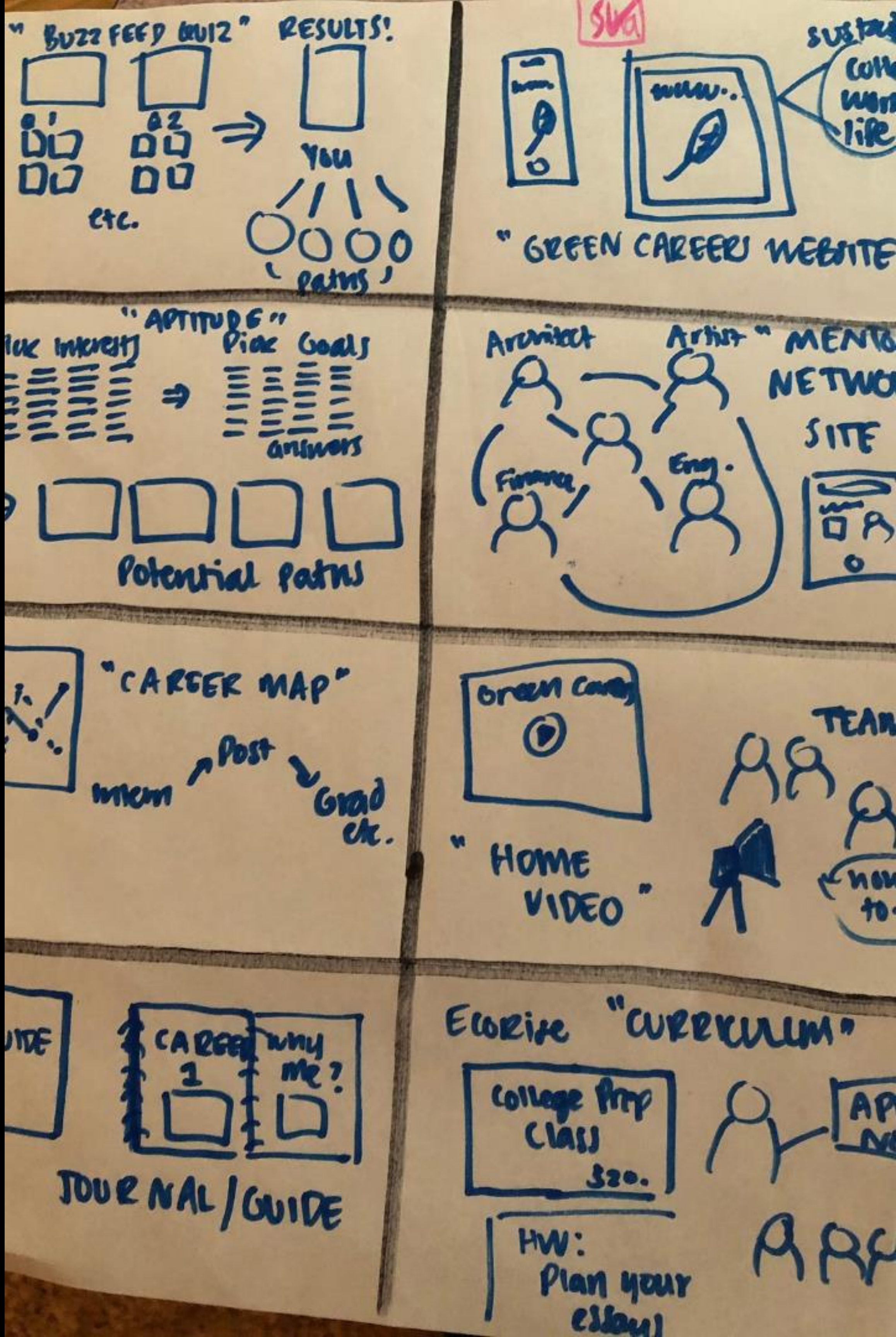
Crazy 8s is a fast-paced technique designed to generate many ideas in a short amount of time. It's commonly used in design sprints and innovation workshops to encourage creativity and overcome mental blocks. The goal is to quickly sketch 8 distinct ideas in 8 minutes.



Crazy 8s:

Step-by-Step

1. Preparation; Pen & Paper
2. Define the Problem or Goal
3. The 8-Minute Timer
4. Set the Rules
5. Evaluate & Prioritise
6. Action Plan



Brainwriting

IDEATE IN SILENCE

Brainwriting is a structured brainstorming method where participants silently write down ideas instead of speaking. Ideas are shared by passing them around, allowing others to build on them. This encourages equal participation, reduces bias and diverse ideas.

Brainwriting:

Step-by-Step

1. Preparation
2. Define the Problem or Goal
3. Give each participant a sheet of paper.
4. Everyone silently writes down 3 ideas in 5 minutes.
5. Pass the paper to the next person.
6. Read the ideas and add or improve them.
7. Repeat for several rounds.
8. Evaluate & Prioritise
9. Action Plan

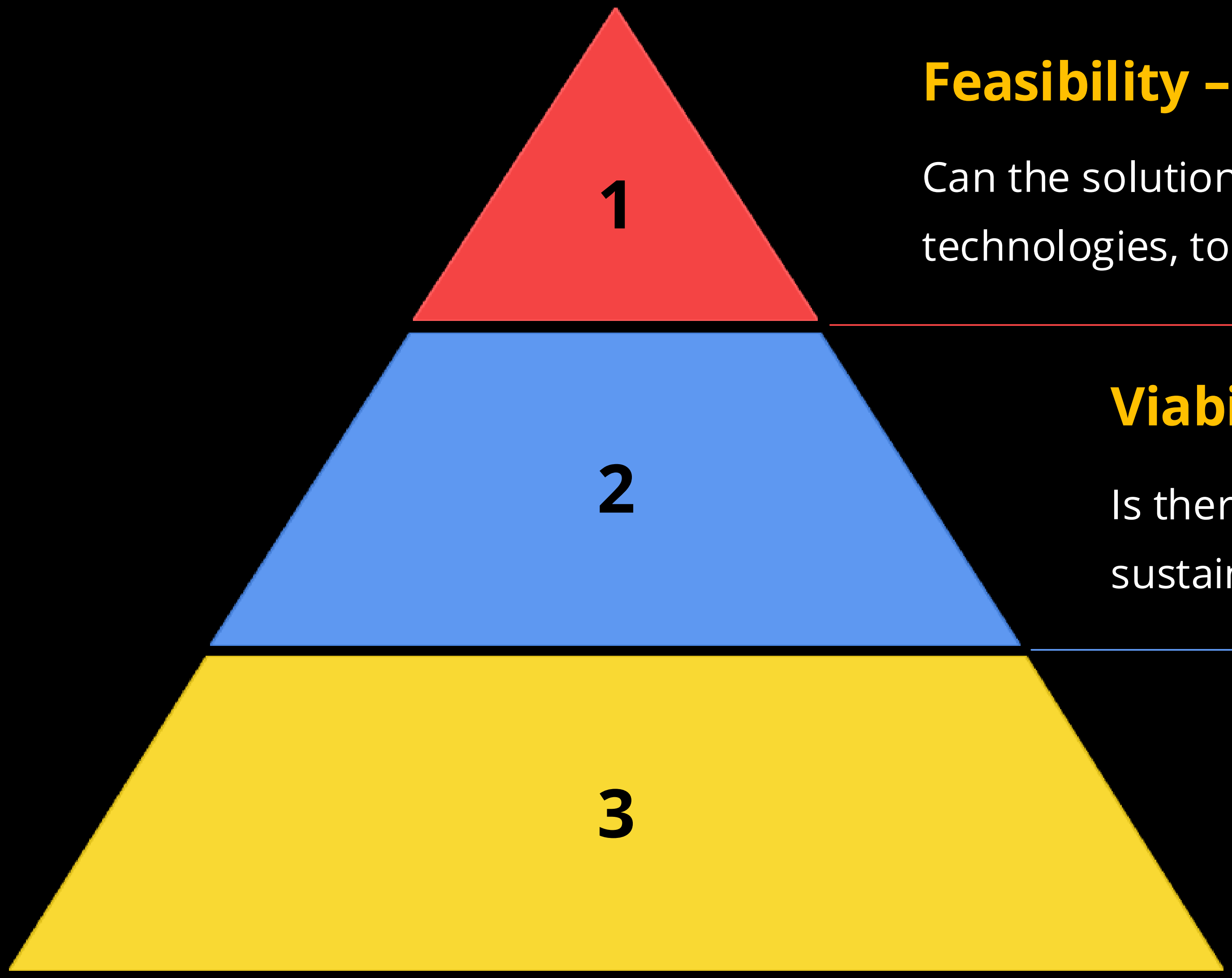
Prompting Gemini

HARNESS THE POWER OF AI

Input all the ideas generated by the team into Gemini to analyse the pros and cons of each idea.

What's the criteria?

Evaluating & Prioritising Ideas



Feasibility – Can We Build This?

Can the solution be technically implemented with the available technologies, tools, skillsets and resources?

Viability – Can We Sell This?

Is there a commercial value for the solution, and can it be sustained financially if turned into a business?

Desirability – Anyone Want This?

Does the solution address real user needs and provide a compelling value proposition?

Prototyping & Testing Solutions

Sketching

Creating low-fidelity prototypes to visualize ideas and communicate concepts effectively to the team.

Wireframing

Developing interactive prototypes to test user flows and interactions, providing a detailed view of the solution.

Prototyping

Building functional prototypes to test user experience and gather real-world feedback.

Incorporate User Feedback & Iteration

1. Analyse

Reviewing and interpreting user feedback, identifying key insights and areas for fixes and improvement.

2. Iterate

Refining the solution based on user feedback, incorporating changes and improvements to enhance the UX.

2. Retest

Re-testing the solution with users to validate improvements and gather more feedback.

Pitfalls to
Avoid in Ideation

Fixation on the First Idea You Got

Avoid settling on the first idea that comes to mind. Explore a variety of options before making a decision.

Premature Evaluation

Don't judge your ideas too quickly. Let the ideation process unfold before critically evaluating the ideas.

Lack of Diversity in Perspectives

Encourage diverse perspectives within the team to avoid group-think and blind spots.

Disregarding Constraints

Consider real-world limitations such as time and technical feasibility.

“ *Your AI tool is only as good as the problem it solves. Don't fall in love with the technology; fall in love with the solution's impact* “

What is LMC?

The Lean Model Canvas is a one-page strategic planning tool used by startups and innovators to quickly map out, test and refine a business idea. Adapted from the traditional Business Model Canvas by Ash Maurya, it focuses on the key assumptions that determine whether a venture can succeed, including the problem being solved, target customers, unique value proposition, solution, revenue model and cost structure.

Its purpose is to help founders validate ideas faster, identify risks early and iterate based on real market feedback instead of spending months building something customers may not want.

LEAN MODEL CANVAS

Map your idea.
Validate fast.
Build what matters.

PROBLEM

List your customer's top 3 problems.

EXISTING ALTERNATIVES

List how these problems are solved today.

SOLUTION

Outline a possible solution for each problem.

KEY METRICS

List the key numbers that tell you how your business is doing.

UNIQUE VALUE PROPOSITION

Single, clear, compelling message that turns an unaware visitor into an interested prospect.

HIGH-LEVEL CONCEPT

List your X for Y analogy (e.g. YouTube = Flickr for videos).

UNFAIR ADVANTAGE

Something that can not be easily copied or bought.

CHANNELS

List your path to customers.

CUSTOMER SEGMENTS

List your target customers and users.

EARLY ADOPTERS

List the characteristics of your ideal customers.

COST STRUCTURE

List your fixed and variable costs.

REVENUE STREAMS

List your sources of revenue.

What Next?

Ideate an innovative AI product that can revolutionise any industry in Malaysia. Pick an industry of your choice.

Use a Lean Model Canvas to pitch your Ideation

(15 mins)